In [68]:
```python
from rdflib import Graph
from rdflib.namespace import RDF
import sys

def clean_ttl_file(input_file: str, output_file: str):
    # Load the RDF graph from the input file
    g = Graph()
    g.parse(input_file, format="turtle")

    # Create a new graph to write the cleaned version
    cleaned_graph = Graph()

    # Group all triples by subject and predicate
    grouped_triples = {}
    for s, p, o in g:
        grouped_triples.setdefault((s, p), set()).add(o)

    # Add grouped triples back into a new graph
    for (s, p), objects in grouped_triples.items():
        for o in objects:
            cleaned_graph.add((s, p, o))

    # Serialize to Turtle with cleaned output
    cleaned_graph.serialize(destination=output_file, format="turtle")

    print(f"Cleaned TTL written to {output_file}")

# Example usage
if __name__ == "__main__":
    input_ttl = "Pro_Go_gan_file.ttl"   # Replace with your actual TTL file
    output_ttl = "Pro_Go_gan_cleaned.ttl"
    clean_ttl_file(input_ttl, output_ttl)
```

```
Cleaned TTL written to Pro_Go_gan_cleaned.ttl
```

In [77]:
```python
from rdflib import Graph, Namespace, URIRef, Literal
import pandas as pd

# Load your RDF file
g = Graph()
g.parse("Pro_Go_gan_Results.ttl", format="turtle")

# Define namespaces
RDF = Namespace("http://www.w3.org/1999/02/22-rdf-syntax-ns#")
oboInOwl = Namespace("http://www.geneontology.org/formats/oboInOwl##")
rdfs = Namespace("http://www.w3.org/2000/01/rdf-schema#")
owl = Namespace("http://www.w3.org/2002/07/owl#")
PR = Namespace("http://purl.obolibrary.org/obo/pr#")

# Bind namespaces for pretty QName output (optional)
g.bind("rdf", RDF)
g.bind("oboInOwl", oboInOwl)
g.bind("rdfs", rdfs)
g.bind("owl", owl)
g.bind("pr", PR)
```

```python
# Old and new base URI strings for replacement
OLD_BASE = "http://example.org/default#"
NEW_BASE = "https://proconsortium.org/cgi-bin/entry_pro?id="

# Properties to extract
properties_to_extract = [
    owl.annotatedProperty,
    owl.annotatedSource,
    rdfs.comment,
    oboInOwl.hasBroadSynonym,
    oboInOwl.hasDbXref,
    oboInOwl.hasExactSynonym,
    oboInOwl.hasNarrowSynonym,
    oboInOwl.hasRelatedSynonym,
    rdfs.label,
    owl.onProperty,
    owl.someValuesFrom,
    rdfs.subClassOf,
    rdfs.type
]

data = []

# Loop over all protein sample subjects
for subj in g.subjects(predicate=RDF.type, object=None):
    subj_str = str(subj)
    if "PR_" in subj_str:  # crude filter for protein samples
        # Replace old base URI with new OBO base URI
        if subj_str.startswith(OLD_BASE):
            protein_sample_uri = subj_str.replace(OLD_BASE, NEW_BASE)
        else:
            protein_sample_uri = subj_str

        entry = {"protein_sample": protein_sample_uri}

        # Extract properties
        for prop in properties_to_extract:
            values = [str(o) for o in g.objects(subj, prop)]

            # Flatten comma-separated string literals if any
            flat = []
            for v in values:
                if isinstance(v, str) and "," in v:
                    flat.extend([item.strip() for item in v.split(",")])
                else:
                    flat.append(v.strip())

            entry[str(prop)] = flat
            entry[str(prop) + "_count"] = len(flat)

        data.append(entry)

# Create DataFrame
df = pd.DataFrame(data)
```

```python
# Optional: show all columns
pd.set_option('display.max_columns', None)
print(df.head())
```

```
                                       protein_sample  \
0  https://proconsortium.org/cgi-bin/entry_pro?id...
1  https://proconsortium.org/cgi-bin/entry_pro?id...
2  https://proconsortium.org/cgi-bin/entry_pro?id...
3  https://proconsortium.org/cgi-bin/entry_pro?id...
4  https://proconsortium.org/cgi-bin/entry_pro?id...

  http://www.w3.org/2002/07/owl#annotatedProperty  \
0                                              []
1                                              []
2                                              []
3                                              []
4                                              []

   http://www.w3.org/2002/07/owl#annotatedProperty_count  \
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

  http://www.w3.org/2002/07/owl#annotatedSource  \
0                                            []
1                                            []
2                                            []
3                                            []
4                                            []

   http://www.w3.org/2002/07/owl#annotatedSource_count  \
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

  http://www.w3.org/2000/01/rdf-schema#comment  \
0                                          []
1                                          []
2                                          []
3                                          []
4                                          []

   http://www.w3.org/2000/01/rdf-schema#comment_count  \
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

  http://www.geneontology.org/formats/oboInOwl##hasBroadSynonym  \
0                                                  []
1                                                  []
2                                                  []
3                                                  []
4                                                  []
```

	http://www.geneontology.org/formats/oboInOwl##hasBroadSynonym_count	http://www.geneontology.org/formats/oboInOwl##hasDbXref	http://www.geneontology.org/formats/oboInOwl##hasDbXref_count	http://www.geneontology.org/formats/oboInOwl##hasExactSynonym	http://www.geneontology.org/formats/oboInOwl##hasExactSynonym_count	http://www.geneontology.org/formats/oboInOwl##hasNarrowSynonym	http://www.geneontology.org/formats/oboInOwl##hasNarrowSynonym_count	http://www.geneontology.org/formats/oboInOwl##hasRelatedSynonym
0	0	[]	0	[]	0	[]	0	[]
1	0	[]	0	[]	0	[]	0	[]
2	0	[]	0	[]	0	[]	0	[]
3	0	[]	0	[]	0	[]	0	[]
4	0	[]	0	[]	0	[]	0	[]

	http://www.geneontology.org/formats/oboInOwl##hasRelatedSynonym_count	http://www.w3.org/2000/01/rdf-schema#label	http://www.w3.org/2000/01/rdf-schema#label_count	http://www.w3.org/2002/07/owl#onProperty	http://www.w3.org/2002/07/owl#onProperty_count	http://www.w3.org/2002/07/owl#someValuesFrom	http://www.w3.org/2002/07/owl#someValuesFrom_count	http://www.w3.org/2000/01/rdf-schema#subClassOf
0	0	[]	0	[]	0	[]	0	[]
1	0	[]	0	[]	0	[]	0	[]
2	0	[]	0	[]	0	[]	0	[]
3	0	[]	0	[]	0	[]	0	[]
4	0	[]	0	[]	0	[]	0	[]

```
   http://www.w3.org/2000/01/rdf-schema#subClassOf_count  \
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

  http://www.w3.org/2000/01/rdf-schema#type  \
0                                        []
1                                        []
2                                        []
3                                        []
4                                        []

   http://www.w3.org/2000/01/rdf-schema#type_count
0                                                0
1                                                0
2                                                0
3                                                0
4                                                0
```

In [71]:
```python
import re

# Extract base ID from full sample URI
def extract_protein_base_id(uri):
    match = re.search(r"PR_\d+", uri)
    return match.group(0) if match else None

# Add the base ID to the DataFrame
df["protein_base_id"] = df["protein_sample"].apply(extract_protein_base_id)
```

In [72]:
```python
grouped = df.groupby("protein_base_id")
```

In [73]:
```python
print(grouped.head())
```

```
                                protein_sample  \
0     http://example.org/default#PR_000000453_1
1     http://example.org/default#PR_000000453_2
2     http://example.org/default#PR_000000453_3
3     http://example.org/default#PR_000000453_4
4     http://example.org/default#PR_000000453_5
...                                         ...
1945  http://example.org/default#PR_000085667_1
1946  http://example.org/default#PR_000085667_2
1947  http://example.org/default#PR_000085667_3
1948  http://example.org/default#PR_000085667_4
1949  http://example.org/default#PR_000085667_5

     http://www.w3.org/2002/07/owl#annotatedProperty  \
0                                                 []
1                                                 []
2                                                 []
3                                                 []
4                                                 []
...                                              ...
1945                                              []
1946                                              []
1947                                              []
1948                                              []
1949                                              []

      http://www.w3.org/2002/07/owl#annotatedProperty_count  \
0                                                         0
1                                                         0
2                                                         0
3                                                         0
4                                                         0
...                                                     ...
1945                                                      0
1946                                                      0
1947                                                      0
1948                                                      0
1949                                                      0

     http://www.w3.org/2002/07/owl#annotatedSource  \
0                                               []
1                                               []
2                                               []
3                                               []
4                                               []
...                                            ...
1945                                            []
1946                                            []
1947                                            []
1948                                            []
1949                                            []

      http://www.w3.org/2002/07/owl#annotatedSource_count  \
0                                                         0
1                                                         0
2                                                         0
```

```
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.w3.org/2000/01/rdf-schema#comment  \
0                                                  []
1                                                  []
2                                                  []
3                                                  []
4                                                  []
...                                                ...
1945                                               []
1946                                               []
1947                                               []
1948                                               []
1949                                               []

      http://www.w3.org/2000/01/rdf-schema#comment_count  \
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.geneontology.org/formats/oboInOwl##hasBroadSynonym  \
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                  ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.geneontology.org/formats/oboInOwl##hasBroadSynonym_count  \
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
```

```
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.geneontology.org/formats/oboInOwl##hasDbXref  \
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.geneontology.org/formats/oboInOwl##hasDbXref_count  \
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.geneontology.org/formats/oboInOwl##hasExactSynonym  \
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.geneontology.org/formats/oboInOwl##hasExactSynonym_count  \
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0
```

```
     http://www.geneontology.org/formats/oboInOwl##hasNarrowSynonym  \
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.geneontology.org/formats/oboInOwl##hasNarrowSynonym_count
\
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.geneontology.org/formats/oboInOwl##hasRelatedSynonym  \
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.geneontology.org/formats/oboInOwl##hasRelatedSynonym_count
\
0                                                     0
1                                                     0
2                                                     0
3                                                     0
4                                                     0
...                                                 ...
1945                                                  0
1946                                                  0
1947                                                  0
1948                                                  0
1949                                                  0

     http://www.w3.org/2000/01/rdf-schema#label  \
```

```
0                                                    []
1                                                    []
2                                                    []
3                                                    []
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.w3.org/2000/01/rdf-schema#label_count  \
0                                                    0
1                                                    0
2                                                    0
3                                                    0
4                                                    0
...                                                ...
1945                                                 0
1946                                                 0
1947                                                 0
1948                                                 0
1949                                                 0

     http://www.w3.org/2002/07/owl#onProperty  \
0                                           []
1                                           []
2                                           []
3                                           []
4                                           []
...                                        ...
1945                                        []
1946                                        []
1947                                        []
1948                                        []
1949                                        []

      http://www.w3.org/2002/07/owl#onProperty_count  \
0                                                    0
1                                                    0
2                                                    0
3                                                    0
4                                                    0
...                                                ...
1945                                                 0
1946                                                 0
1947                                                 0
1948                                                 0
1949                                                 0

     http://www.w3.org/2002/07/owl#someValuesFrom  \
0                                              []
1                                              []
2                                              []
3                                              []
```

```
4                                                    []
...                                                 ...
1945                                                 []
1946                                                 []
1947                                                 []
1948                                                 []
1949                                                 []

      http://www.w3.org/2002/07/owl#someValuesFrom_count  \
0                                                      0
1                                                      0
2                                                      0
3                                                      0
4                                                      0
...                                                  ...
1945                                                   0
1946                                                   0
1947                                                   0
1948                                                   0
1949                                                   0

     http://www.w3.org/2000/01/rdf-schema#subClassOf  \
0                                                  []
1                                                  []
2                                                  []
3                                                  []
4                                                  []
...                                               ...
1945                                               []
1946                                               []
1947                                               []
1948                                               []
1949                                               []

      http://www.w3.org/2000/01/rdf-schema#subClassOf_count  \
0                                                         0
1                                                         0
2                                                         0
3                                                         0
4                                                         0
...                                                     ...
1945                                                      0
1946                                                      0
1947                                                      0
1948                                                      0
1949                                                      0

     http://www.w3.org/2000/01/rdf-schema#type  \
0                                           []
1                                           []
2                                           []
3                                           []
4                                           []
...                                        ...
1945                                        []
1946                                        []
```

```
1947                                                []
1948                                                []
1949                                                []

      http://www.w3.org/2000/01/rdf-schema#type_count protein_base_id
0                                                   0    PR_000000453
1                                                   0    PR_000000453
2                                                   0    PR_000000453
3                                                   0    PR_000000453
4                                                   0    PR_000000453
...                                               ...             ...
1945                                                0    PR_000085667
1946                                                0    PR_000085667
1947                                                0    PR_000085667
1948                                                0    PR_000085667
1949                                                0    PR_000085667

[1950 rows x 28 columns]
```

In [74]:

```python
import pandas as pd

properties_to_extract = [
    owl.annotatedProperty,
    owl.annotatedSource,
    rdfs.comment,
    oboInOwl.hasBroadSynonym,
    oboInOwl.hasDbXref,
    oboInOwl.hasExactSynonym,
    oboInOwl.hasNarrowSynonym,
    oboInOwl.hasRelatedSynonym,
    rdfs.label,
    owl.onProperty,
    owl.someValuesFrom,
    rdfs.subClassOf,
    rdfs.type
]

# Build aggregation dict dynamically for each property URI string with the "
agg_dict = {str(prop) + '_count': ['mean', 'std', 'min', 'max'] for prop in

# Then aggregate
summary = grouped.agg(agg_dict)

print(summary.head())
```

```
                http://www.w3.org/2002/07/owl#annotatedProperty_count
\
                                                                  mean  std
protein_base_id
PR_000000453                                                       0.0     0.0
PR_000000843                                                       0.0     0.0
PR_000000992                                                       0.0     0.0
PR_000001167                                                       0.0     0.0
PR_000001705                                                       0.0     0.0

                       http://www.w3.org/2002/07/owl#annotatedSource_count
\
                min max                                                 mean
protein_base_id
PR_000000453      0   0                                                 0.0
PR_000000843      0   0                                                 0.0
PR_000000992      0   0                                                 0.0
PR_000001167      0   0                                                 0.0
PR_000001705      0   0                                                 0.0

                                \
                 std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.w3.org/2000/01/rdf-schema#comment_count
\
                                                          mean  std min
protein_base_id
PR_000000453                                               0.0  0.0   0
PR_000000843                                               0.0  0.0   0
PR_000000992                                               0.0  0.0   0
PR_000001167                                               0.0  0.0   0
PR_000001705                                               0.0  0.0   0

                    \
                max
protein_base_id
PR_000000453      0
PR_000000843      0
PR_000000992      0
PR_000001167      0
PR_000001705      0

                http://www.geneontology.org/formats/oboInOwl##hasBroadSynony
m_count  \

mean
protein_base_id
PR_000000453                                               0.0
PR_000000843                                               0.0
PR_000000992                                               0.0
```

```
PR_000001167                                                    0.0
PR_000001705                                                    0.0

                                \
                 std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.geneontology.org/formats/oboInOwl##hasDbXref_coun
t  \
                                                                      mea
n
protein_base_id
PR_000000453                                                    0.0
PR_000000843                                                    0.0
PR_000000992                                                    0.0
PR_000001167                                                    0.0
PR_000001705                                                    0.0

                                \
                 std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.geneontology.org/formats/oboInOwl##hasExactSynony
m_count  \

mean
protein_base_id
PR_000000453                                                    0.0
PR_000000843                                                    0.0
PR_000000992                                                    0.0
PR_000001167                                                    0.0
PR_000001705                                                    0.0

                                \
                 std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.geneontology.org/formats/oboInOwl##hasNarrowSynon
ym_count  \

mean
```

```
protein_base_id
PR_000000453                                                    0.0
PR_000000843                                                    0.0
PR_000000992                                                    0.0
PR_000001167                                                    0.0
PR_000001705                                                    0.0

                                \
                 std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.geneontology.org/formats/oboInOwl##hasRelatedSyno
nym_count  \

mean
protein_base_id
PR_000000453                                                    0.0
PR_000000843                                                    0.0
PR_000000992                                                    0.0
PR_000001167                                                    0.0
PR_000001705                                                    0.0

                              http://www.w3.org/2000/01/rdf-schema#label_coun
t  \
                 std min max                                                mea
n
protein_base_id
PR_000000453     0.0   0   0                                                 0.
0
PR_000000843     0.0   0   0                                                 0.
0
PR_000000992     0.0   0   0                                                 0.
0
PR_000001167     0.0   0   0                                                 0.
0
PR_000001705     0.0   0   0                                                 0.
0

                              http://www.w3.org/2002/07/owl#onProperty_count
\
                 std min max                                               mean
protein_base_id
PR_000000453     0.0   0   0                                                0.0
PR_000000843     0.0   0   0                                                0.0
PR_000000992     0.0   0   0                                                0.0
PR_000001167     0.0   0   0                                                0.0
PR_000001705     0.0   0   0                                                0.0

                                \
                 std min max
protein_base_id
```

```
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0

                http://www.w3.org/2002/07/owl#someValuesFrom_count
\
                                                                 mean  std min
protein_base_id
PR_000000453                                                      0.0  0.0   0
PR_000000843                                                      0.0  0.0   0
PR_000000992                                                      0.0  0.0   0
PR_000001167                                                      0.0  0.0   0
PR_000001705                                                      0.0  0.0   0

                    http://www.w3.org/2000/01/rdf-schema#subClassOf_count  \
                max                                                   mean
protein_base_id
PR_000000453      0                                                  0.0
PR_000000843      0                                                  0.0
PR_000000992      0                                                  0.0
PR_000001167      0                                                  0.0
PR_000001705      0                                                  0.0

                             http://www.w3.org/2000/01/rdf-schema#type_count
\
                  std min max                                          mean
protein_base_id
PR_000000453     0.0   0   0                                          0.0
PR_000000843     0.0   0   0                                          0.0
PR_000000992     0.0   0   0                                          0.0
PR_000001167     0.0   0   0                                          0.0
PR_000001705     0.0   0   0                                          0.0


                  std min max
protein_base_id
PR_000000453     0.0   0   0
PR_000000843     0.0   0   0
PR_000000992     0.0   0   0
PR_000001167     0.0   0   0
PR_000001705     0.0   0   0
```

In [75]:

```python
# Flatten multi-index columns created by agg()
summary.columns = ['_'.join(col).strip() for col in summary.columns.values]

# Reset index for easier plotting and handling
summary = summary.reset_index()

# Melt the dataframe to long format focusing on 'mean' statistics
melted = summary.melt(
    id_vars=['protein_base_id'],  # keep the group identifier column(s)
    value_vars=[col for col in summary.columns if '_mean' in col],  # only m
    var_name='property_stat',
    value_name='value'
```

```
)

# Extract property name only (remove the trailing '_mean')
melted['property'] = melted['property_stat'].apply(lambda x: x.replace('_mea

print(melted.head())
```

```
  protein_base_id                                      property_stat  value
\
0    PR_000000453  http://www.w3.org/2002/07/owl#annotatedPropert...    0.0
1    PR_000000843  http://www.w3.org/2002/07/owl#annotatedPropert...    0.0
2    PR_000000992  http://www.w3.org/2002/07/owl#annotatedPropert...    0.0
3    PR_000001167  http://www.w3.org/2002/07/owl#annotatedPropert...    0.0
4    PR_000001705  http://www.w3.org/2002/07/owl#annotatedPropert...    0.0

                                          property
0  http://www.w3.org/2002/07/owl#annotatedPropert...
1  http://www.w3.org/2002/07/owl#annotatedPropert...
2  http://www.w3.org/2002/07/owl#annotatedPropert...
3  http://www.w3.org/2002/07/owl#annotatedPropert...
4  http://www.w3.org/2002/07/owl#annotatedPropert...
```

In [66]:

```
import plotly.express as px

# Encode protein_base_id as categorical numeric for X-axis
melted['protein_num'] = melted['protein_base_id'].astype('category').cat.coc

# Encode property as categorical numeric for Y-axis
melted['property_num'] = melted['property'].astype('category').cat.codes

fig = px.scatter_3d(
    melted,
    x='protein_num',
    y='property_num',
    z='value',
    color='protein_base_id',    # Color by protein_base_id
    symbol='property',          # Different symbol for each property
    labels={
        'protein_num': 'Protein Base ID',
        'property_num': 'Property',
        'value': 'Mean Count'
    },
    title='3D Distribution of Property Counts per Protein'
)

# Update axis ticks to show categorical labels instead of numeric codes
fig.update_layout(
    scene=dict(
        xaxis=dict(
            tickmode='array',
            tickvals=melted['protein_num'].unique(),
            ticktext=melted['protein_base_id'].astype('category').cat.categc
        ),
        yaxis=dict(
            tickmode='array',
            tickvals=melted['property_num'].unique(),
```

```
            ticktext=melted['property'].astype('category').cat.categories
        )
    )
)

fig.show()
```

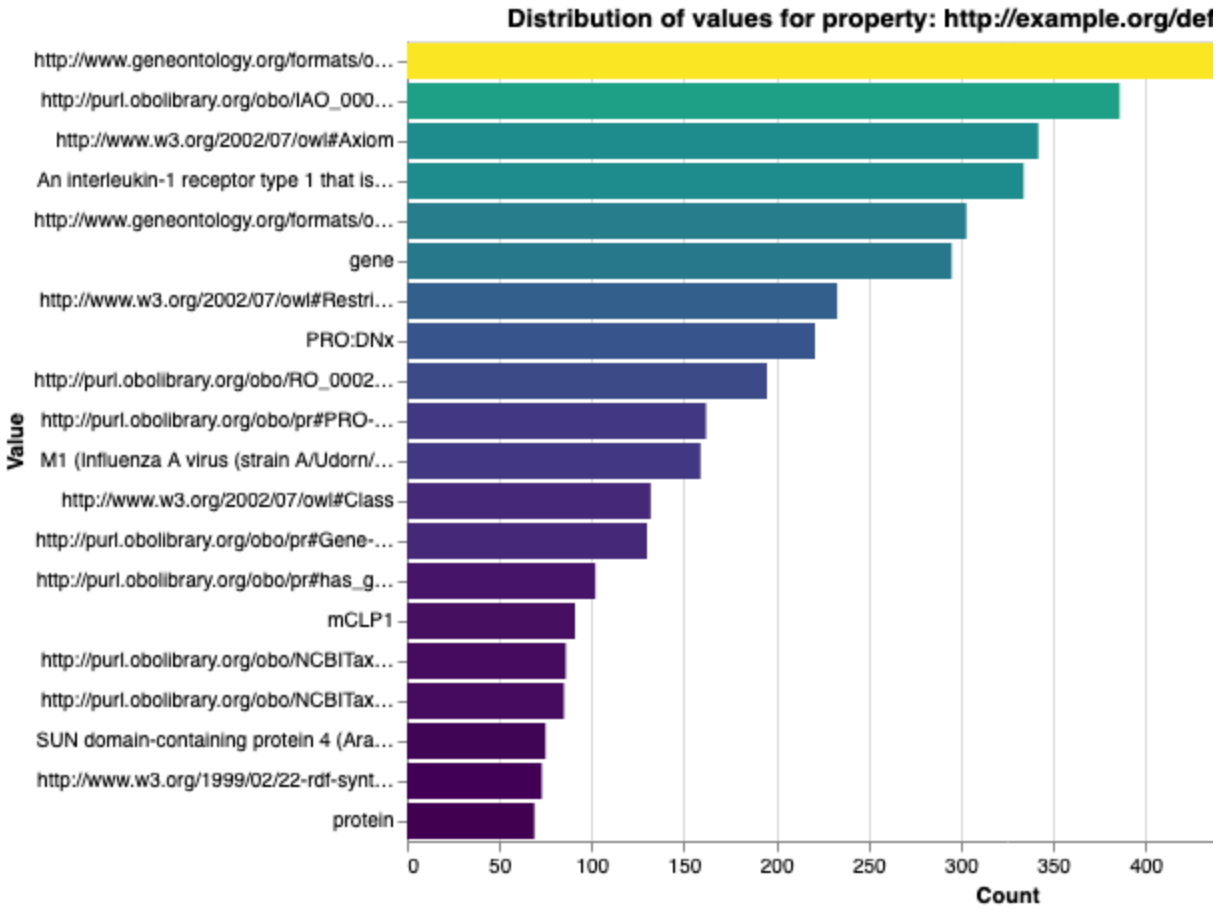
In [76]:

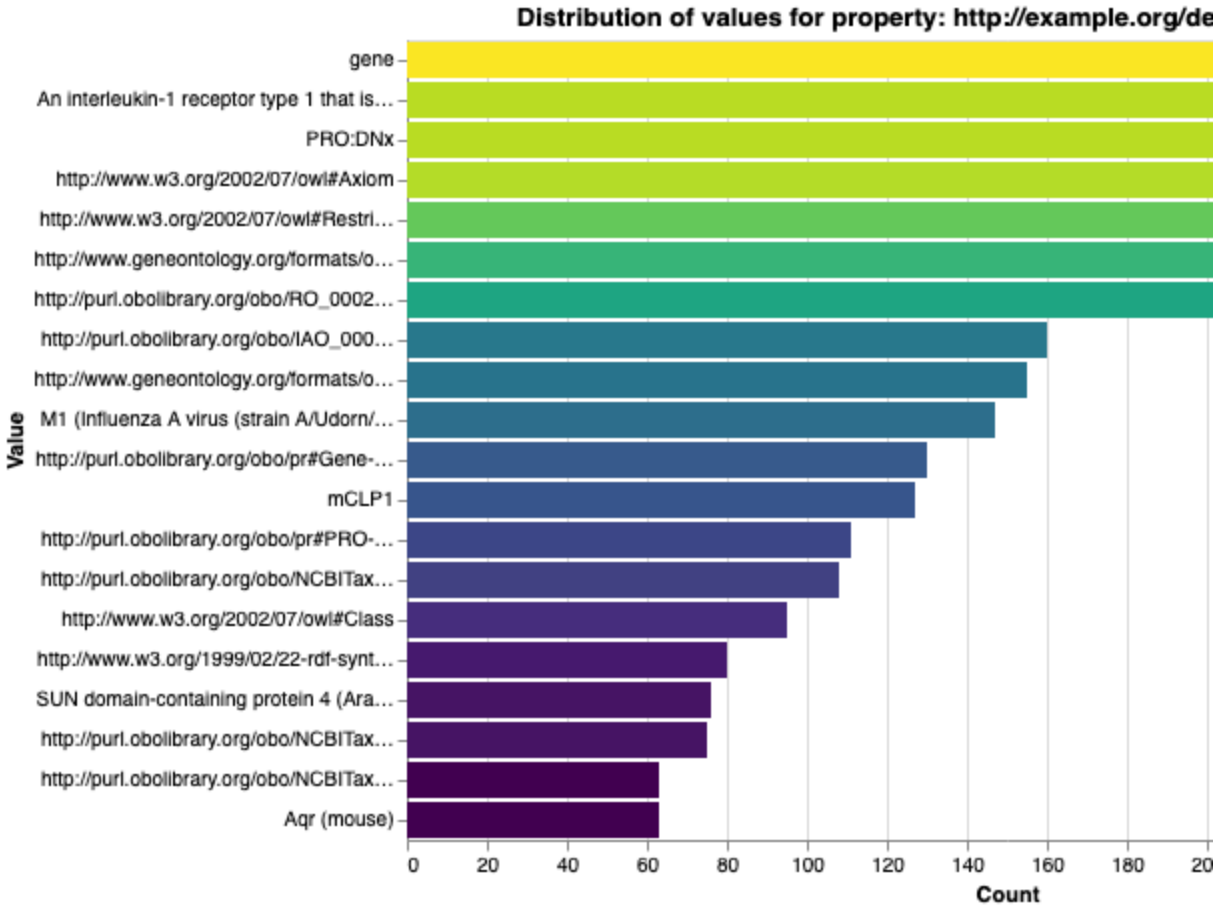
```
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter

# List of property names as strings (keys in df)
property_keys = [str(p) for p in properties_to_extract]

# For each property, aggregate all values across samples
for prop in property_keys:
    # Extract all lists of values for this property in all rows
    all_values_lists = df[prop].dropna().tolist()

    # Flatten list of lists into one big list of values
    all_values = [val for sublist in all_values_lists for val in sublist]

    if len(all_values) == 0:
        print(f"No data for property {prop}")
        continue

    # Count frequency of each distinct value
    value_counts = Counter(all_values)

    # Convert to DataFrame for plotting
    count_df = pd.DataFrame(value_counts.items(), columns=['Value', 'Count']
    count_df = count_df.sort_values('Count', ascending=False).head(20)  # To

    # Plot
    plt.figure(figsize=(10,6))
    sns.barplot(data=count_df, x='Count', y='Value', palette='viridis')
    plt.title(f'Distribution of values for property:\n{prop}')
    plt.xlabel('Count')
    plt.ylabel('Value')
    plt.tight_layout()
    plt.show()
```

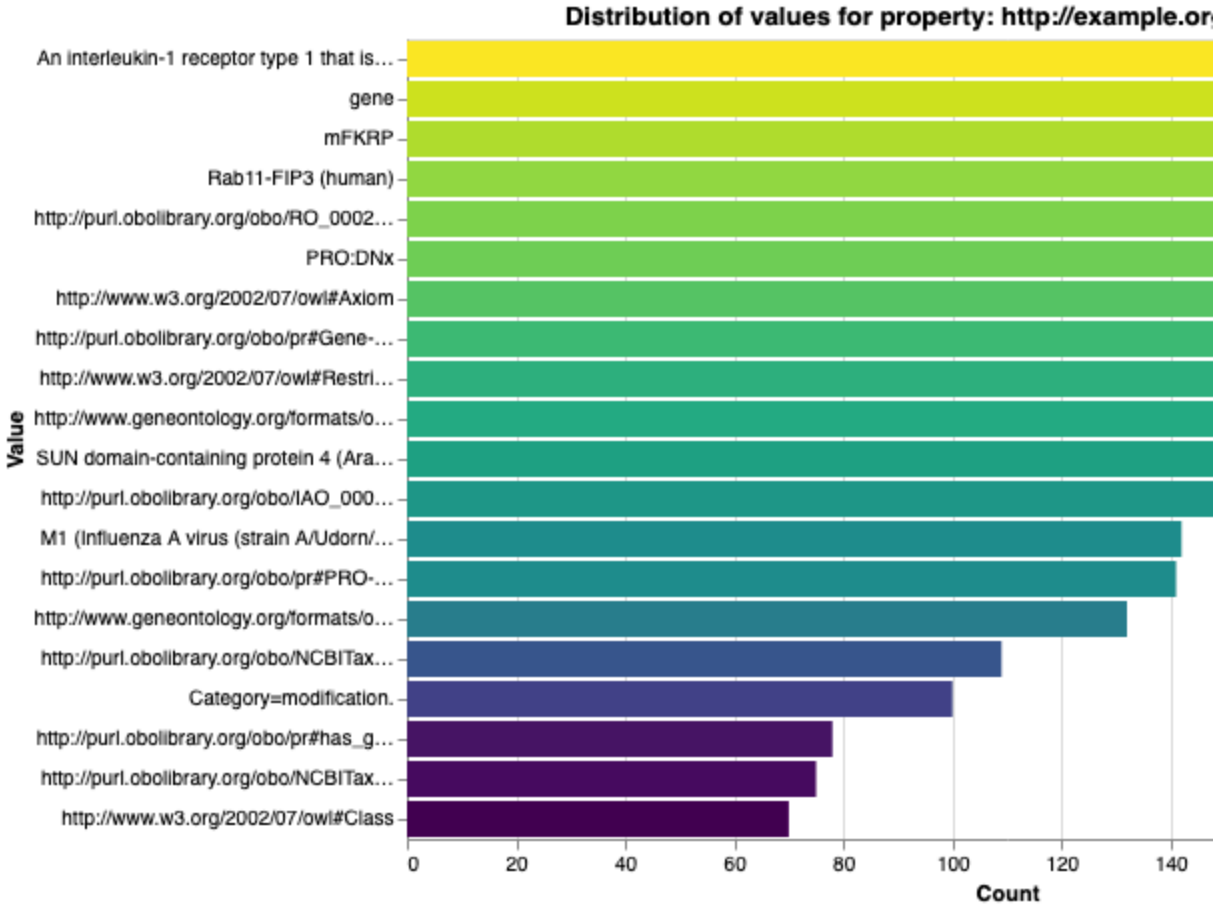
```
No data for property http://www.w3.org/2002/07/owl#annotatedProperty
No data for property http://www.w3.org/2002/07/owl#annotatedSource
No data for property http://www.w3.org/2000/01/rdf-schema#comment
No data for property http://www.geneontology.org/formats/oboInOwl##hasBroadS
ynonym
No data for property http://www.geneontology.org/formats/oboInOwl##hasDbXref
No data for property http://www.geneontology.org/formats/oboInOwl##hasExactS
ynonym
No data for property http://www.geneontology.org/formats/oboInOwl##hasNarrow
Synonym
No data for property http://www.geneontology.org/formats/oboInOwl##hasRelate
dSynonym
No data for property http://www.w3.org/2000/01/rdf-schema#label
No data for property http://www.w3.org/2002/07/owl#onProperty
No data for property http://www.w3.org/2002/07/owl#someValuesFrom
No data for property http://www.w3.org/2000/01/rdf-schema#subClassOf
No data for property http://www.w3.org/2000/01/rdf-schema#type
```

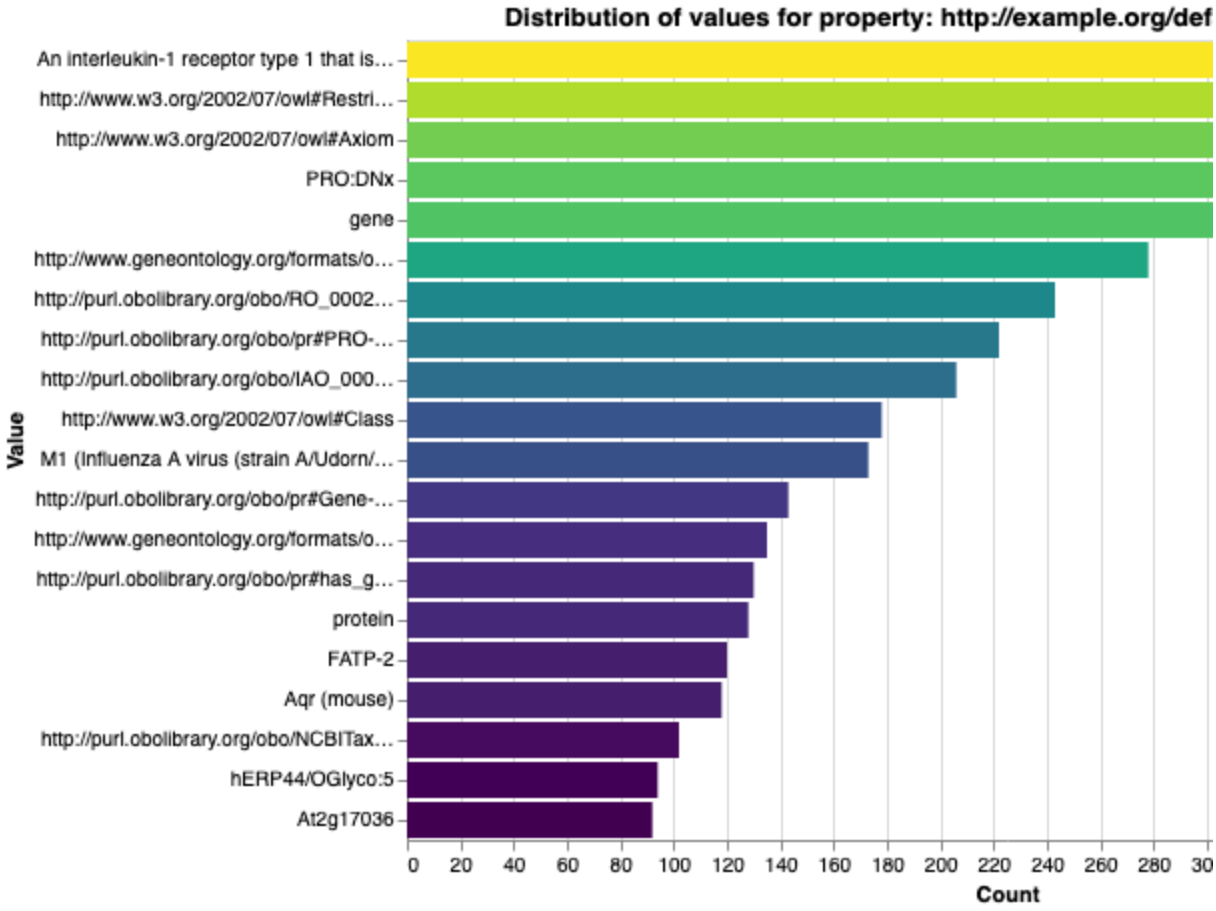
In [23]:

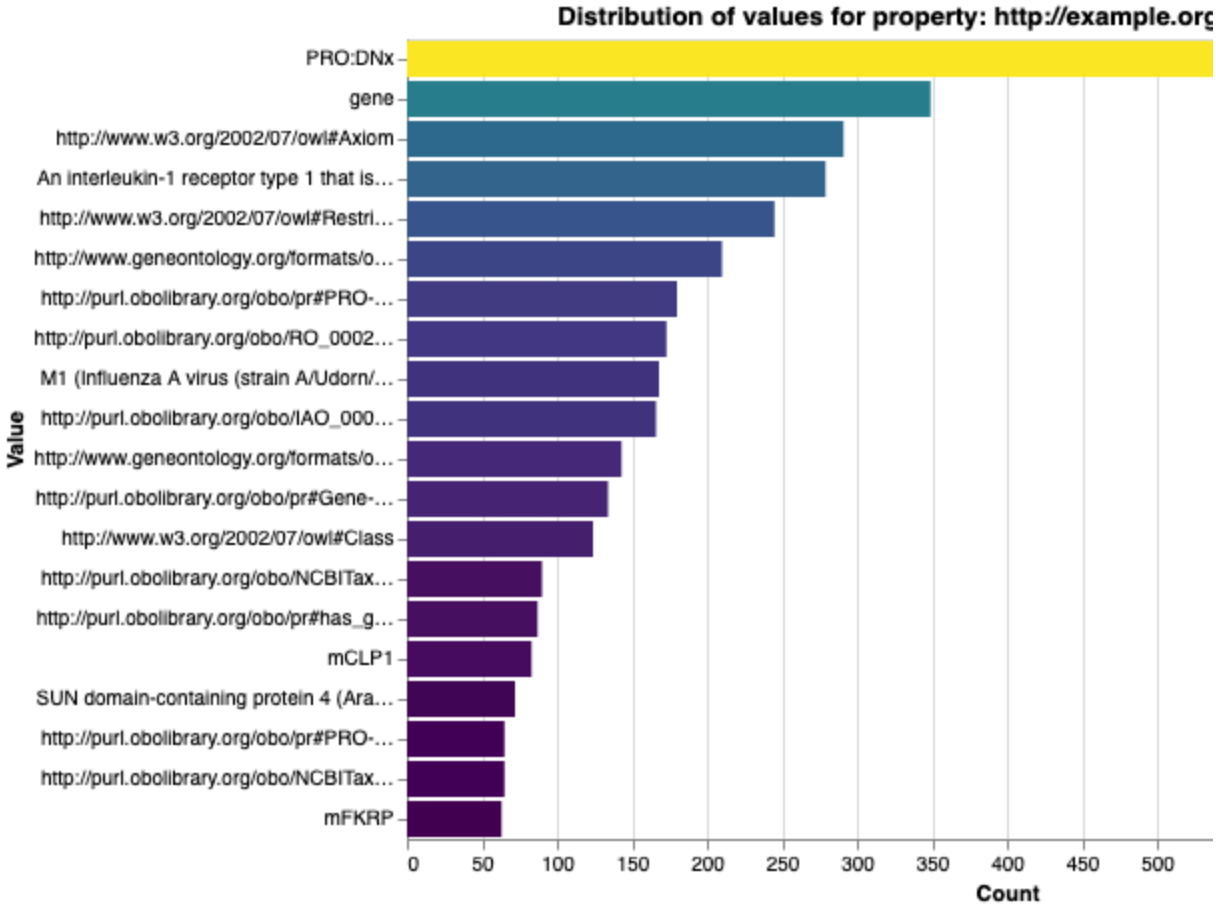
```python
import altair as alt
import pandas as pd
from collections import Counter

alt.renderers.enable('default')

property_keys = [str(p) for p in properties_to_extract]

for prop in property_keys:
    all_values_lists = df[prop].dropna().tolist()
    all_values = [val for sublist in all_values_lists for val in sublist]

    if len(all_values) == 0:
        print(f"No data for property {prop}")
        continue

    value_counts = Counter(all_values)
    count_df = pd.DataFrame(value_counts.items(), columns=['Value', 'Count']
    count_df = count_df.sort_values('Count', ascending=False).head(20)  # to

    chart = alt.Chart(count_df).mark_bar().encode(
        y=alt.Y('Value:N', sort='-x', title='Value'),
        x=alt.X('Count:Q', title='Count'),
        color=alt.Color('Count:Q', scale=alt.Scale(scheme='viridis'), legenc
        tooltip=['Value', 'Count']
    ).properties(
        title=f'Distribution of values for property:\n{prop}',
        width=600,
        height=400
    )

    chart.display()
```

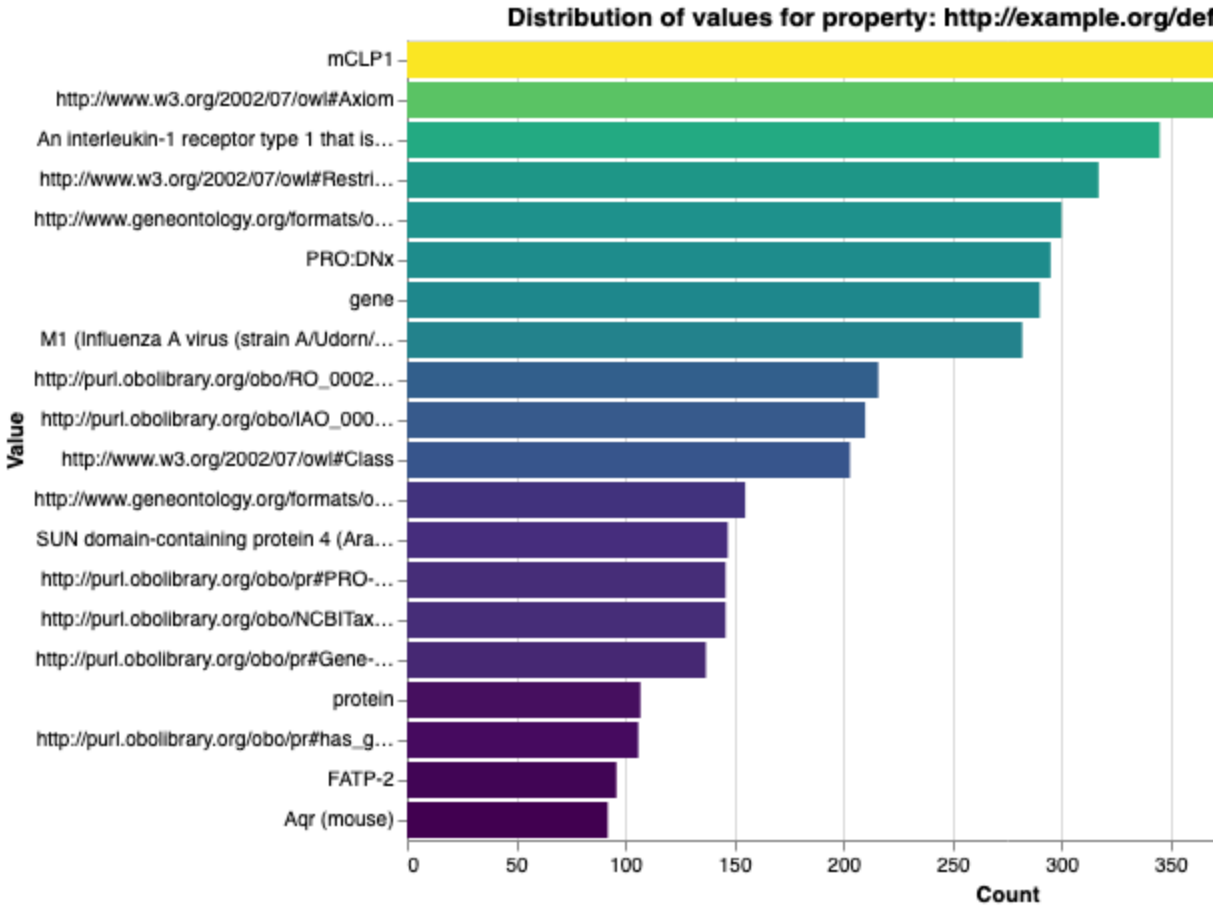

Distribution of values for property: http://example.org/def
http://www.geneontology.org/formats/o…
http://purl.obolibrary.org/obo/IAO_000…
http://www.w3.org/2002/07/owl#Axiom
An interleukin-1 receptor type 1 that is…
http://www.geneontology.org/formats/o…
gene
http://www.w3.org/2002/07/owl#Restri…
PRO:DNx
http://purl.obolibrary.org/obo/RO_0002…
http://purl.obolibrary.org/obo/pr#PRO-…
M1 (Influenza A virus (strain A/Udorn/…
http://www.w3.org/2002/07/owl#Class
http://purl.obolibrary.org/obo/pr#Gene-…
http://purl.obolibrary.org/obo/pr#has_g…
mCLP1
http://purl.obolibrary.org/obo/NCBITax…
http://purl.obolibrary.org/obo/NCBITax…
SUN domain-containing protein 4 (Ara…
http://www.w3.org/1999/02/22-rdf-synt…
protein
Value
0
50
100
150
200
250
300
350
400
Count

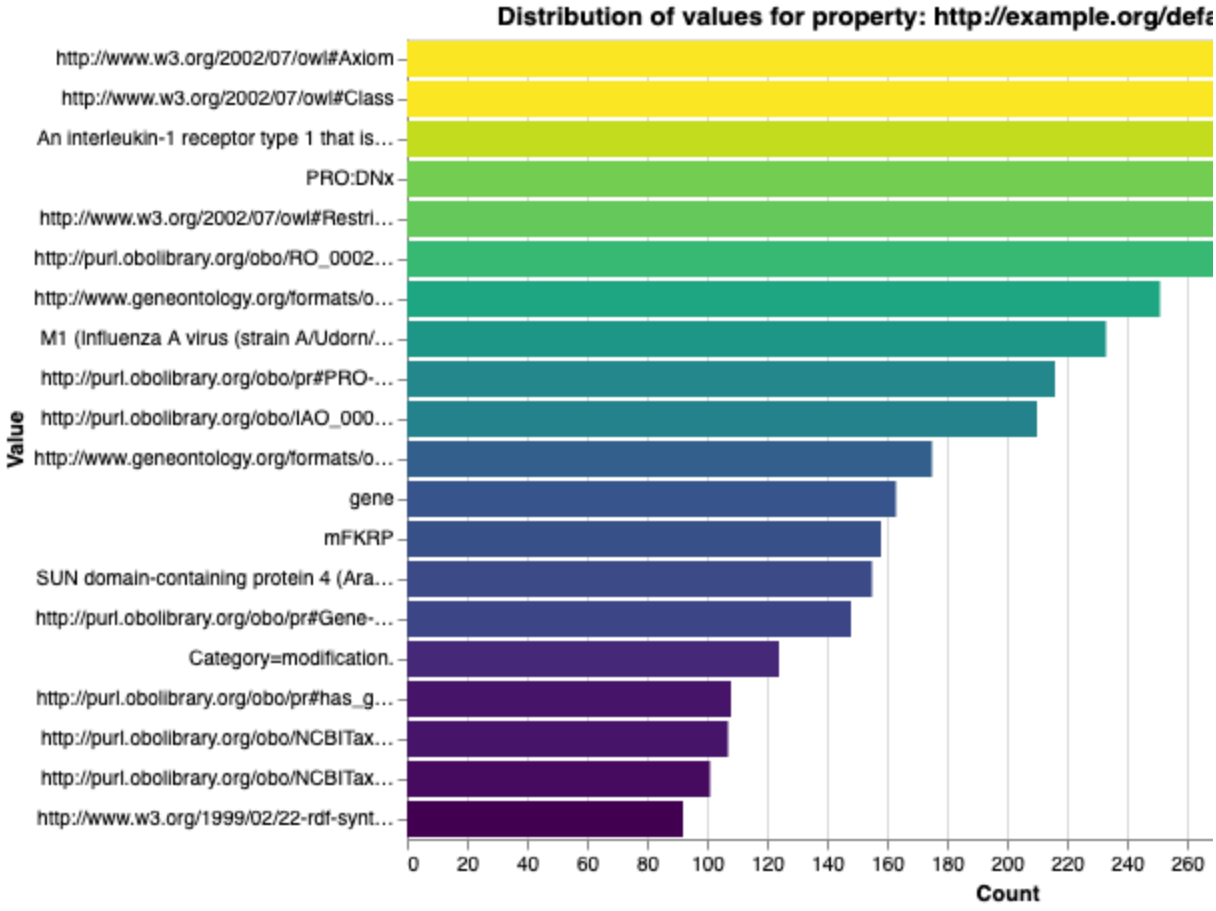

Distribution of values for property: http://example.org/de
gene
An interleukin-1 receptor type 1 that is…
PRO:DNx
http://www.w3.org/2002/07/owl#Axiom
http://www.w3.org/2002/07/owl#Restri…
http://www.geneontology.org/formats/o…
http://purl.obolibrary.org/obo/RO_0002…
http://purl.obolibrary.org/obo/IAO_000…
http://www.geneontology.org/formats/o…
M1 (Influenza A virus (strain A/Udorn/…
http://purl.obolibrary.org/obo/pr#Gene-…
mCLP1
http://purl.obolibrary.org/obo/pr#PRO-…
http://purl.obolibrary.org/obo/NCBITax…
http://www.w3.org/2002/07/owl#Class
http://www.w3.org/1999/02/22-rdf-synt…
SUN domain-containing protein 4 (Ara…
http://purl.obolibrary.org/obo/NCBITax…
http://purl.obolibrary.org/obo/NCBITax…
Aqr (mouse)
Value
0
20
40
60
80
100
120
140
160
180
20
Count

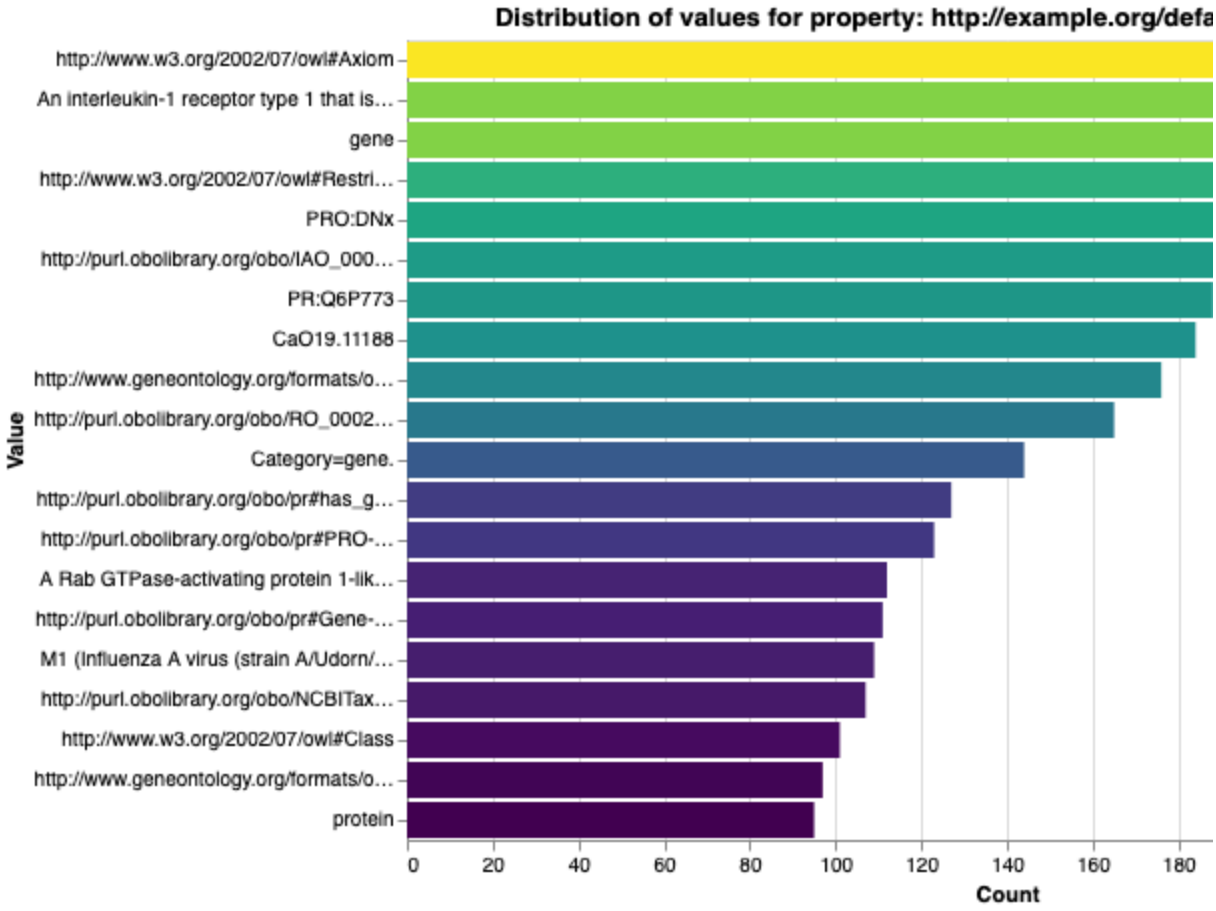

Distribution of values for property: http://example.or
An interleukin-1 receptor type 1 that is...
gene
mFKRP
Rab11-FIP3 (human)
http://purl.obolibrary.org/obo/RO_0002...
PRO:DNx
http://www.w3.org/2002/07/owl#Axiom
http://purl.obolibrary.org/obo/pr#Gene-...
http://www.w3.org/2002/07/owl#Restri...
http://www.geneontology.org/formats/o...
SUN domain-containing protein 4 (Ara...
http://purl.obolibrary.org/obo/IAO_000...
M1 (Influenza A virus (strain A/Udorn/...
http://purl.obolibrary.org/obo/pr#PRO-...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/NCBITax...
Category=modification.
http://purl.obolibrary.org/obo/pr#has_g...
http://purl.obolibrary.org/obo/NCBITax...
http://www.w3.org/2002/07/owl#Class
Value
0
20
40
60
80
100
120
140
Count

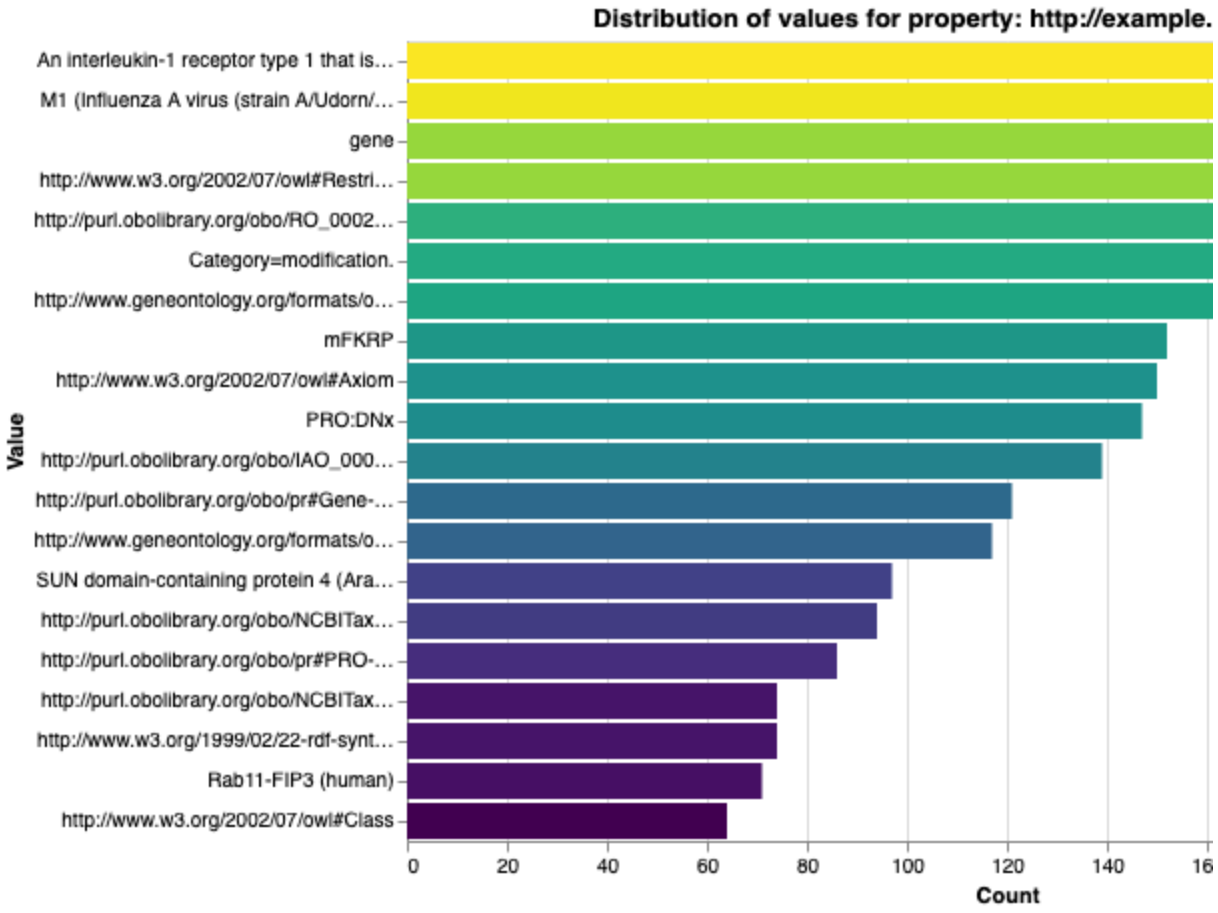

Distribution of values for property: http://example.org/def
An interleukin-1 receptor type 1 that is...
http://www.w3.org/2002/07/owl#Restri...
http://www.w3.org/2002/07/owl#Axiom
PRO:DNx
gene
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/RO_0002...
http://purl.obolibrary.org/obo/pr#PRO-...
http://purl.obolibrary.org/obo/IAO_000...
http://www.w3.org/2002/07/owl#Class
M1 (Influenza A virus (strain A/Udorn/...
http://purl.obolibrary.org/obo/pr#Gene-...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/pr#has_g...
protein
FATP-2
Aqr (mouse)
http://purl.obolibrary.org/obo/NCBITax...
hERP44/OGlyco:5
At2g17036
Value
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
30
Count

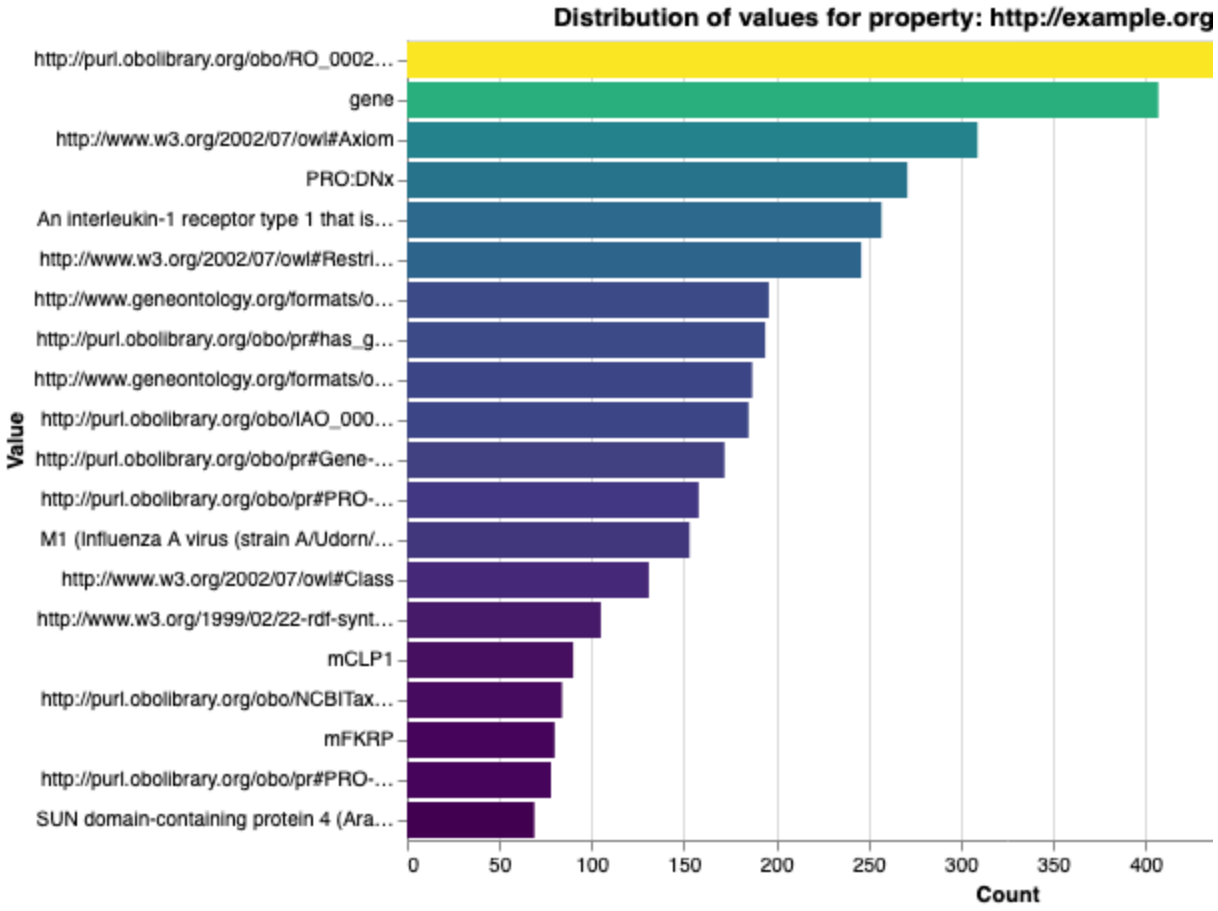

Distribution of values for property: http://example.org
PRO:DNx
gene
http://www.w3.org/2002/07/owl#Axiom
An interleukin-1 receptor type 1 that is…
http://www.w3.org/2002/07/owl#Restri…
http://www.geneontology.org/formats/o…
http://purl.obolibrary.org/obo/pr#PRO-…
http://purl.obolibrary.org/obo/RO_0002…
M1 (Influenza A virus (strain A/Udorn/…
http://purl.obolibrary.org/obo/IAO_000…
http://www.geneontology.org/formats/o…
http://purl.obolibrary.org/obo/pr#Gene-…
http://www.w3.org/2002/07/owl#Class
http://purl.obolibrary.org/obo/NCBITax…
http://purl.obolibrary.org/obo/pr#has_g…
mCLP1
SUN domain-containing protein 4 (Ara…
http://purl.obolibrary.org/obo/pr#PRO-…
http://purl.obolibrary.org/obo/NCBITax…
mFKRP
Value
0
50
100
150
200
250
300
350
400
450
500
Count

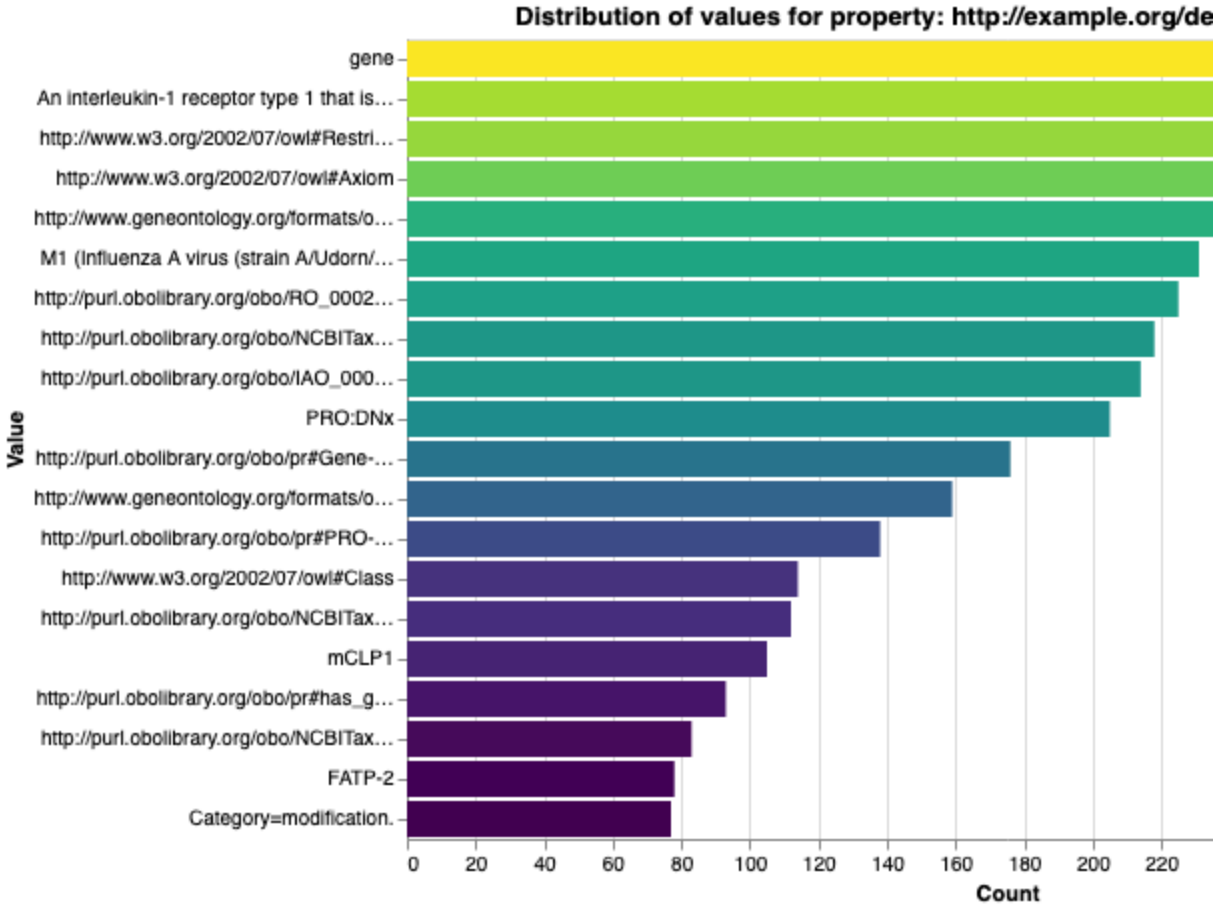

Distribution of values for property: http://example.org/def
mCLP1
http://www.w3.org/2002/07/owl#Axiom
An interleukin-1 receptor type 1 that is…
http://www.w3.org/2002/07/owl#Restri…
http://www.geneontology.org/formats/o…
PRO:DNx
gene
M1 (Influenza A virus (strain A/Udorn/…
http://purl.obolibrary.org/obo/RO_0002…
http://purl.obolibrary.org/obo/IAO_000…
http://www.w3.org/2002/07/owl#Class
http://www.geneontology.org/formats/o…
SUN domain-containing protein 4 (Ara…
http://purl.obolibrary.org/obo/pr#PRO-…
http://purl.obolibrary.org/obo/NCBITax…
http://purl.obolibrary.org/obo/pr#Gene-…
protein
http://purl.obolibrary.org/obo/pr#has_g…
FATP-2
Aqr (mouse)
Value
0
50
100
150
200
250
300
350
Count

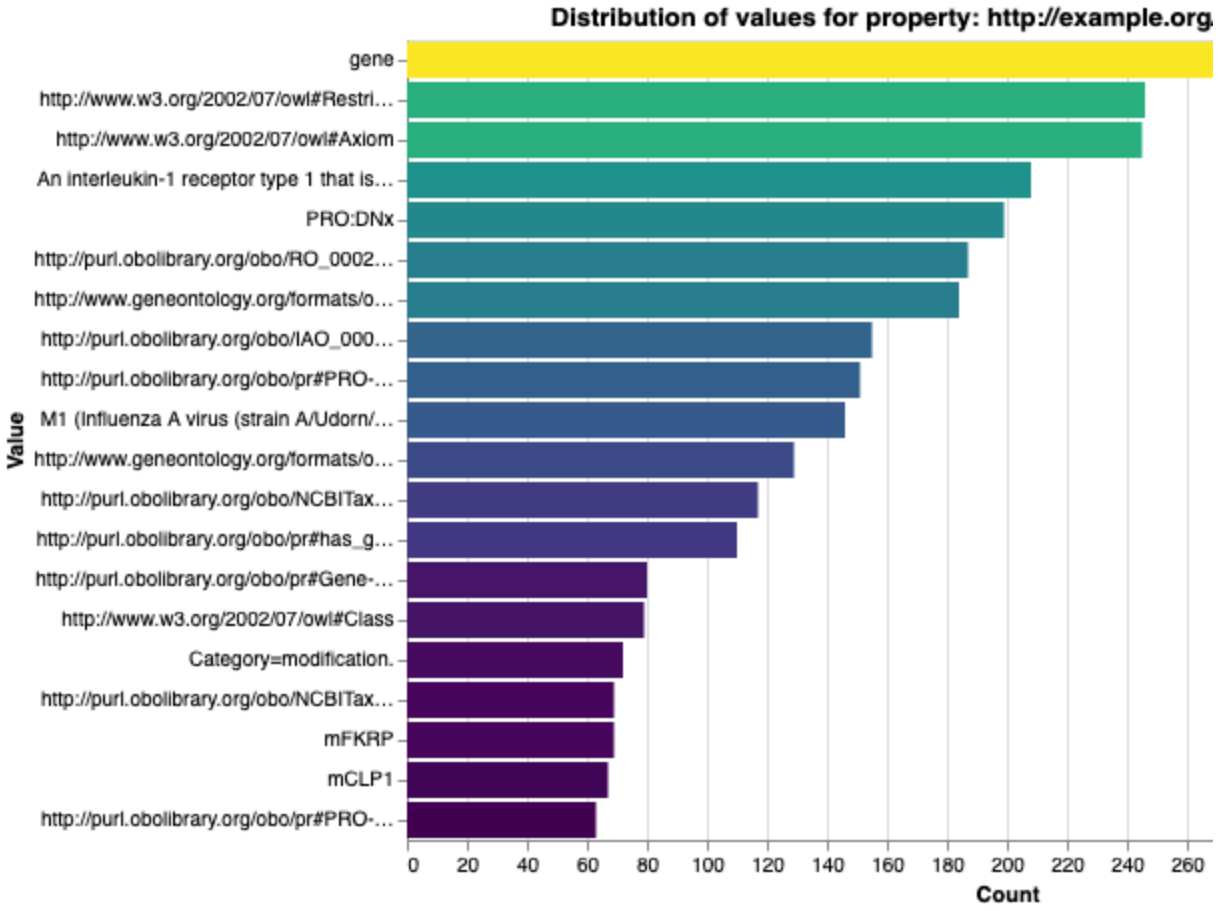

Distribution of values for property: http://example.org/defa
http://www.w3.org/2002/07/owl#Axiom
http://www.w3.org/2002/07/owl#Class
An interleukin-1 receptor type 1 that is…
PRO:DNx
http://www.w3.org/2002/07/owl#Restri…
http://purl.obolibrary.org/obo/RO_0002…
http://www.geneontology.org/formats/o…
M1 (Influenza A virus (strain A/Udorn/…
http://purl.obolibrary.org/obo/pr#PRO-…
http://purl.obolibrary.org/obo/IAO_000…
http://www.geneontology.org/formats/o…
gene
mFKRP
SUN domain-containing protein 4 (Ara…
http://purl.obolibrary.org/obo/pr#Gene-…
Category=modification.
http://purl.obolibrary.org/obo/pr#has_g…
http://purl.obolibrary.org/obo/NCBITax…
http://purl.obolibrary.org/obo/NCBITax…
http://www.w3.org/1999/02/22-rdf-synt…
Value
0 20 40 60 80 100 120 140 160 180 200 220 240 260
Count

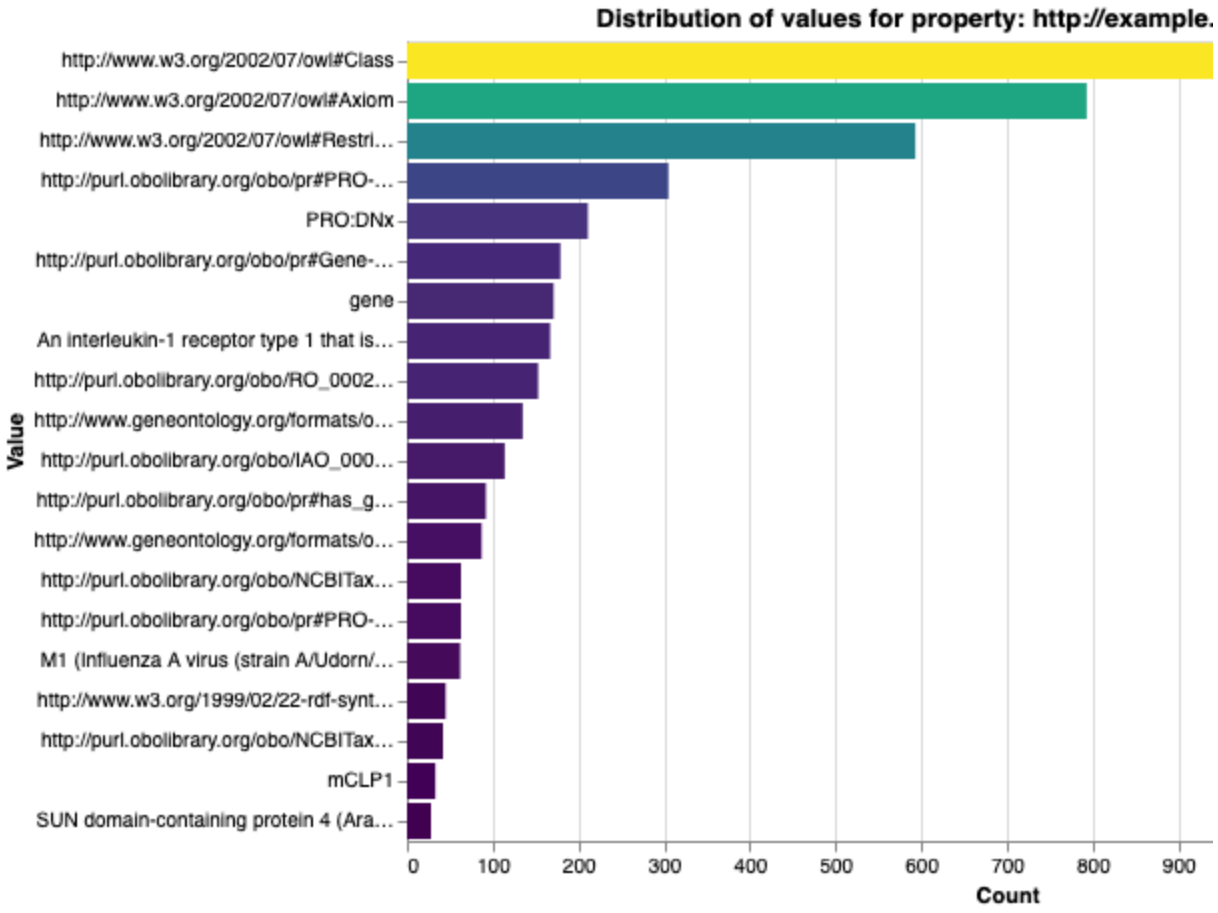

Distribution of values for property: http://example.org/defa
http://www.w3.org/2002/07/owl#Axiom
An interleukin-1 receptor type 1 that is…
gene
http://www.w3.org/2002/07/owl#Restri…
PRO:DNx
http://purl.obolibrary.org/obo/IAO_000…
PR:Q6P773
CaO19.11188
http://www.geneontology.org/formats/o…
http://purl.obolibrary.org/obo/RO_0002…
Category=gene.
http://purl.obolibrary.org/obo/pr#has_g…
http://purl.obolibrary.org/obo/pr#PRO-…
A Rab GTPase-activating protein 1-lik…
http://purl.obolibrary.org/obo/pr#Gene-…
M1 (Influenza A virus (strain A/Udorn/…
http://purl.obolibrary.org/obo/NCBITax…
http://www.w3.org/2002/07/owl#Class
http://www.geneontology.org/formats/o…
protein
Value
0 20 40 60 80 100 120 140 160 180
Count


Distribution of values for property: http://example.
An interleukin-1 receptor type 1 that is...
M1 (Influenza A virus (strain A/Udorn/...
gene
http://www.w3.org/2002/07/owl#Restri...
http://purl.obolibrary.org/obo/RO_0002...
Category=modification.
http://www.geneontology.org/formats/o...
mFKRP
http://www.w3.org/2002/07/owl#Axiom
PRO:DNx
http://purl.obolibrary.org/obo/IAO_000...
http://purl.obolibrary.org/obo/pr#Gene-...
http://www.geneontology.org/formats/o...
SUN domain-containing protein 4 (Ara...
http://purl.obolibrary.org/obo/NCBITax...
http://purl.obolibrary.org/obo/pr#PRO-...
http://purl.obolibrary.org/obo/NCBITax...
http://www.w3.org/1999/02/22-rdf-synt...
Rab11-FIP3 (human)
http://www.w3.org/2002/07/owl#Class
Value
0
20
40
60
80
100
120
140
Count


Distribution of values for property: http://example.org
http://purl.obolibrary.org/obo/RO_0002...
gene
http://www.w3.org/2002/07/owl#Axiom
PRO:DNx
An interleukin-1 receptor type 1 that is...
http://www.w3.org/2002/07/owl#Restri...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/pr#has_g...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/IAO_000...
http://purl.obolibrary.org/obo/pr#Gene-...
http://purl.obolibrary.org/obo/pr#PRO-...
M1 (Influenza A virus (strain A/Udorn/...
http://www.w3.org/2002/07/owl#Class
http://www.w3.org/1999/02/22-rdf-synt...
mCLP1
http://purl.obolibrary.org/obo/NCBITax...
mFKRP
http://purl.obolibrary.org/obo/pr#PRO-...
SUN domain-containing protein 4 (Ara...
Value
0
50
100
150
200
250
300
350
400
Count


Distribution of values for property: http://example.org/de
gene
An interleukin-1 receptor type 1 that is...
http://www.w3.org/2002/07/owl#Restri...
http://www.w3.org/2002/07/owl#Axiom
http://www.geneontology.org/formats/o...
M1 (Influenza A virus (strain A/Udorn/...
http://purl.obolibrary.org/obo/RO_0002...
http://purl.obolibrary.org/obo/NCBITax...
http://purl.obolibrary.org/obo/IAO_000...
PRO:DNx
http://purl.obolibrary.org/obo/pr#Gene-...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/pr#PRO-...
http://www.w3.org/2002/07/owl#Class
http://purl.obolibrary.org/obo/NCBITax...
mCLP1
http://purl.obolibrary.org/obo/pr#has_g...
http://purl.obolibrary.org/obo/NCBITax...
FATP-2
Category=modification.
Value
0 20 40 60 80 100 120 140 160 180 200 220
Count


Distribution of values for property: http://example.org
gene
http://www.w3.org/2002/07/owl#Restri...
http://www.w3.org/2002/07/owl#Axiom
An interleukin-1 receptor type 1 that is...
PRO:DNx
http://purl.obolibrary.org/obo/RO_0002...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/IAO_000...
http://purl.obolibrary.org/obo/pr#PRO-...
M1 (Influenza A virus (strain A/Udorn/...
http://www.geneontology.org/formats/o...
http://purl.obolibrary.org/obo/NCBITax...
http://purl.obolibrary.org/obo/pr#has_g...
http://purl.obolibrary.org/obo/pr#Gene-...
http://www.w3.org/2002/07/owl#Class
Category=modification.
http://purl.obolibrary.org/obo/NCBITax...
mFKRP
mCLP1
http://purl.obolibrary.org/obo/pr#PRO-...
Value
0 20 40 60 80 100 120 140 160 180 200 220 240 260
Count



In [43]:

```python
from rdflib import Graph, Namespace, URIRef, BNode, Literal

# Define namespaces matching your prefixes
NS_MAP = {
    "PR": Namespace("http://purl.obolibrary.org/obo/pr#"),
    "NCBITaxon": Namespace("http://purl.obolibrary.org/obo/NCBITaxon_"),
    "RO": Namespace("http://purl.obolibrary.org/obo/RO_"),
    "oboInOwl": Namespace("http://www.geneontology.org/formats/oboInOwl#"),
    "IAO": Namespace("http://purl.obolibrary.org/obo/IAO_"),
    "ex": Namespace("http://example.org/shapes#"),
    "dcterms": Namespace("http://purl.org/dc/terms/"),
    "sh": Namespace("http://www.w3.org/ns/shacl#"),
    "rdf": Namespace("http://www.w3.org/1999/02/22-rdf-syntax-ns#"),
    "xsd": Namespace("http://www.w3.org/2001/XMLSchema#"),
    "rdfs": Namespace("http://www.w3.org/2000/01/rdf-schema#"),
    "owl": Namespace("http://www.w3.org/2002/07/owl#"),
}

DEFAULT_NS = Namespace("http://example.org/default#")

def replace_ns(uri):
    """If the URI starts with default NS, try to map to a known prefix from
    if not isinstance(uri, URIRef):
        return uri  # Leave literals and BNodes unchanged

    uri_str = str(uri)
    if uri_str.startswith(str(DEFAULT_NS)):
        local_part = uri_str[len(DEFAULT_NS):]
```

```
        # Now try to detect which prefix it should have based on the local p
        # Example: PR_000011340 -> should map to PR
        # Example: NCBITaxon_9606 -> should map to NCBITaxon
        for prefix, ns in NS_MAP.items():
            if local_part.startswith(prefix + "_") or local_part.startswith(
                # Construct new URI with correct namespace + remainder
                # Remove prefix from local_part if it exists
                suffix = local_part[len(prefix):] if local_part.startswith(p
                suffix = suffix.lstrip("_")  # Remove leading underscore if
                return ns[suffix]
        # If no match, fallback to original URI
        return uri
    else:
        return uri

def fix_graph_ns(g):
    new_g = Graph()
    # Bind namespaces for better serialization
    for prefix, ns in NS_MAP.items():
        new_g.bind(prefix, ns)
    new_g.bind("default", DEFAULT_NS)

    for s, p, o in g:
        s_fixed = replace_ns(s)
        p_fixed = replace_ns(p)
        o_fixed = replace_ns(o)
        new_g.add((s_fixed, p_fixed, o_fixed))
    return new_g

if __name__ == "__main__":
    g = Graph()
    g.parse("Pro_Go_vae_Results.ttl", format="ttl")  # Replace with your TTL

    fixed_graph = fix_graph_ns(g)

    fixed_graph.serialize(destination="corrected_vae_file.ttl", format="ttl"
    print("Namespace replacement done. Saved to corrected_file.ttl")
```

```
Namespace replacement done. Saved to corrected_file.ttl
```

In [44]:

```
from rdflib import Graph, Namespace, URIRef, Literal, BNode

# Your known namespaces
NS_MAP = {
    "PR": Namespace("http://purl.obolibrary.org/obo/pr#"),
    "NCBITaxon": Namespace("http://purl.obolibrary.org/obo/NCBITaxon_"),
    "RO": Namespace("http://purl.obolibrary.org/obo/RO_"),
    "oboInOwl": Namespace("http://www.geneontology.org/formats/oboInOwl#"),
    "IAO": Namespace("http://purl.obolibrary.org/obo/IAO_"),
    "ex": Namespace("http://example.org/shapes#"),
    "dcterms": Namespace("http://purl.org/dc/terms/"),
    "sh": Namespace("http://www.w3.org/ns/shacl#"),
    "rdf": Namespace("http://www.w3.org/1999/02/22-rdf-syntax-ns#"),
    "xsd": Namespace("http://www.w3.org/2001/XMLSchema#"),
    "rdfs": Namespace("http://www.w3.org/2000/01/rdf-schema#"),
```

```
    "owl": Namespace("http://www.w3.org/2002/07/owl#"),
    "default": Namespace("http://example.org/default#"),
}

# Map property local names (without prefix) to correct namespace prefix stri
PROPERTY_TO_NS_PREFIX = {
    # from your example TTL mappings:
    "annotatedProperty": "owl",
    "annotatedSource": "owl",
    "comment": "rdfs",
    "hasBroadSynonym": "oboInOwl",
    "hasDbXref": "oboInOwl",
    "hasExactSynonym": "oboInOwl",
    "hasNarrowSynonym": "oboInOwl",
    "hasRelatedSynonym": "oboInOwl",
    "label": "rdfs",
    "onProperty": "owl",
    "someValuesFrom": "owl",
    "subClassOf": "rdfs",
    "type": "rdf",
    "created": "dcterms",
    "date": "dcterms",
    # Add more mappings here if needed
}

def fix_uri(uri):
    """Fix URIRefs that start with default: by replacing with correct NS bas
    if not isinstance(uri, URIRef):
        return uri
    uri_str = str(uri)
    default_ns_str = str(NS_MAP["default"])
    if uri_str.startswith(default_ns_str):
        local_name = uri_str[len(default_ns_str):]
        # If it's a property predicate, fix namespace
        ns_prefix = PROPERTY_TO_NS_PREFIX.get(local_name)
        if ns_prefix and ns_prefix in NS_MAP:
            return NS_MAP[ns_prefix][local_name]
        else:
            # Could be a resource, keep default or handle differently
            # Optionally fallback to default namespace
            return uri
    else:
        return uri

def fix_predicate(pred):
    """Fix predicate URIs with default prefix by mapping local name to corre
    if not isinstance(pred, URIRef):
        return pred
    pred_str = str(pred)
    default_ns_str = str(NS_MAP["default"])
    if pred_str.startswith(default_ns_str):
        local_name = pred_str[len(default_ns_str):]
        ns_prefix = PROPERTY_TO_NS_PREFIX.get(local_name)
        if ns_prefix and ns_prefix in NS_MAP:
            return NS_MAP[ns_prefix][local_name]
        else:
```

```
            # fallback, keep original
            return pred
    else:
        return pred

def fix_object(obj):
    """Fix object URIs similarly if they start with default prefix."""
    if isinstance(obj, URIRef):
        return fix_uri(obj)
    return obj  # literals or blank nodes unchanged

def fix_graph(g):
    new_g = Graph()
    # Bind namespaces for better output
    for prefix, ns in NS_MAP.items():
        new_g.bind(prefix, ns)

    for s, p, o in g:
        p_fixed = fix_predicate(p)
        o_fixed = fix_object(o)
        new_g.add((s, p_fixed, o_fixed))
    return new_g

if __name__ == "__main__":
    g = Graph()
    g.parse("corrected_vae_file.ttl", format="ttl")  # your TTL input file

    fixed_g = fix_graph(g)
    fixed_g.serialize("corrected_file.ttl", format="ttl")
    print("Done! Output saved to corrected_file.ttl")
```

```
Done! Output saved to corrected_file.ttl
```

In [87]:

```
from rdflib import Graph, Namespace, URIRef, Literal
from collections import defaultdict, Counter
import pandas as pd

# Load your RDF data file
g = Graph()
g.parse("Pro_Go_gan_file.ttl", format="turtle")  # <-- Replace with your RDF

# Define namespaces
RDF = Namespace("http://www.w3.org/1999/02/22-rdf-syntax-ns#")
oboInOwl = Namespace("http://www.geneontology.org/formats/oboInOwl#")
rdfs = Namespace("http://www.w3.org/2000/01/rdf-schema#")
owl = Namespace("http://www.w3.org/2002/07/owl#")
PR = Namespace("http://purl.obolibrary.org/obo/pr#")

# Bind namespaces (optional for nicer output)
g.bind("rdf", RDF)
g.bind("oboInOwl", oboInOwl)
g.bind("rdfs", rdfs)
g.bind("owl", owl)
g.bind("pr", PR)

# Old and new base URI strings for protein ID replacement
```

```
OLD_BASE = "http://purl.obolibrary.org/obo/pr#"
NEW_BASE = "https://proconsortium.org/cgi-bin/entry_pro?id="

# Properties to extract (using URIRefs)
properties_to_extract = [
    owl.annotatedProperty,
    owl.annotatedSource,
    rdfs.comment,
    oboInOwl.hasBroadSynonym,
    oboInOwl.hasDbXref,
    oboInOwl.hasExactSynonym,
    oboInOwl.hasNarrowSynonym,
    oboInOwl.hasRelatedSynonym,
    rdfs.label,
    owl.onProperty,
    owl.someValuesFrom,
    rdfs.subClassOf,
    RDF.type
]

# Dictionary to store values per protein_id and property
protein_data = defaultdict(lambda: defaultdict(list))
protein_ids_found = set()

# Iterate over all triples and extract relevant data
for subj, pred, obj in g:
    if pred not in properties_to_extract:
        continue

    subj_str = str(subj)

    # Process only subjects starting with the old protein URI prefix
    if subj_str.startswith(OLD_BASE):
        # Extract the part after the prefix, e.g. "000000453_1"
        sample_part = subj_str[len(OLD_BASE):]

        # Extract base protein ID before underscore (e.g., "000000453")
        base_id = sample_part.split('_')[0]

        protein_ids_found.add(base_id)

        # Create new protein URI with the new base + base_id
        protein_uri = NEW_BASE + base_id

        # Convert object to string (URIRefs or Literals)
        if isinstance(obj, Literal):
            value_str = str(obj)
        elif isinstance(obj, URIRef):
            value_str = str(obj)
        else:
            value_str = str(obj)

        # Append the value grouped by protein and property
        protein_data[protein_uri][pred].append(value_str)

# Print all unique base protein IDs found
```

```python
print(f"Found base protein IDs ({len(protein_ids_found)}): {sorted(protein_i

# Prepare list to hold statistics
stats = []

# Calculate stats per protein and property
for protein, props in protein_data.items():
    for prop, values in props.items():
        # Flatten comma-separated values (some values may be multiple, e.g.
        flat_values = []
        for v in values:
            if ',' in v:
                flat_values.extend([x.strip() for x in v.split(',')])
            else:
                flat_values.append(v)

        counts = Counter(flat_values)
        total = sum(counts.values())
        unique_count = len(counts)
        most_common_val, most_common_count = counts.most_common(1)[0] if uni

        stats.append({
            "protein_id": protein,
            "property": prop.n3(g.namespace_manager),  # Pretty QName like c
            "unique_values_count": unique_count,
            "most_common_value": most_common_val,
            "most_common_count": most_common_count,
            "value_counts": dict(counts),
            "total_values": total
        })

# Create a DataFrame for easy viewing or export
df_stats = pd.DataFrame(stats)

# Display the stats
print(df_stats)

# Optional: Save to CSV if you want
# df_stats.to_csv("protein_property_stats.csv", index=False)
```

Found base protein IDs (1391): ['000000453', '000000843', '000000992', '0000
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[18083 rows x 7 columns]
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In [89]:

```python
import re

def replace_protein_sample_type(ttl_text):
    # regex to match lines like: PR:000000453_1 a PR:000000453,
    pattern = r'(PR:\d+_\d+)\s+a\s+PR:(\d+),'

    def repl(match):
        sample_iri = match.group(1)
        protein_id = match.group(2)
        # replace with PR:<sample> a <https://proconsortium.org/cgi-bin/entr
        return f'{sample_iri} a <https://proconsortium.org/cgi-bin/entry_pro

    replaced_text = re.sub(pattern, repl, ttl_text)
    return replaced_text

# Example usage:
with open('Pro_Go_vae_file.ttl', 'r') as f:
```

```
    ttl_content = f.read()

updated_ttl_content = replace_protein_sample_type(ttl_content)

with open('VAE_results.ttl', 'w') as f:
    f.write(updated_ttl_content)
```